MAIS PAC

IPhO 02.16: Heat Transfer, Phase Transitions and Radiation · Lecture + Problems

Heat Transfer, Phase Transitions and Radiation

MAIS Physics & Astronomy Club (PAC) · IPhO lecture + problem handout

7 lecture sections · 10 problems · 1 interactive experiment · 5 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to olympiad measurements. Five vector diagrams and an interactive experiment support the argument; ten graded problems and independent worked solutions turn the ideas into practice. The coverage checklist is the IPhO syllabus, not this handout.

Contents & learning path

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1. Three ways for heat to move

Thermodynamics says **how much** heat flows; heat transfer says **how fast**. There are exactly three mechanisms:

- **Conduction:** energy passes from molecule to molecule without bulk motion. Dominant in solids.
- **Convection:** a moving fluid carries energy with it. Dominant in liquids and gases, and the reason radiators are at floor level.
- **Radiation:** energy is carried by electromagnetic waves and needs no medium at all. The only mechanism that works through a vacuum.

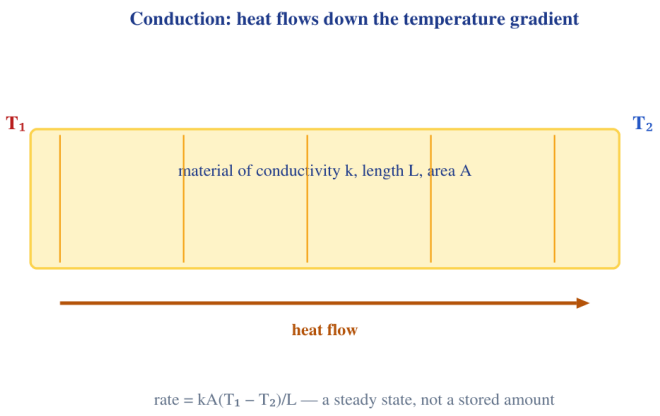


Figure 1. Once the steady state is reached, the same power crosses every cross-section — otherwise energy would pile up. That is why layers of different materials add thermal resistances, exactly like resistors in series.

2. Conduction and thermal resistance

For a slab of area A, thickness d and thermal conductivity k, with faces held at T_1 and T_2, the steady heat current is

$$(dQ)/(dt) = (kA(T_1-T_2))/(d) = (\Delta T)/(R_({th})), R_({th}) = (d)/(kA). \quad (1)$$

This is Ohm's law with temperature in place of voltage: layers in series add their resistances, and the same heat current passes through every layer once the steady state is reached.

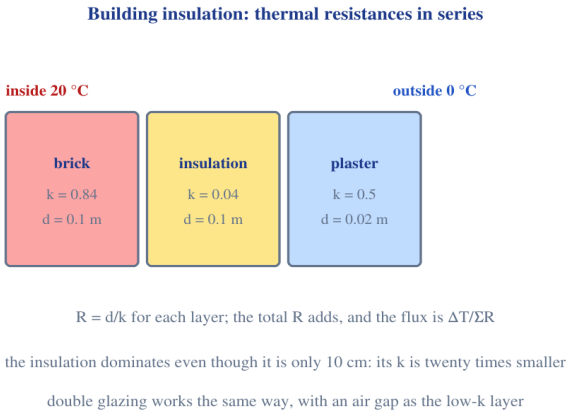


Figure 4. Choosing insulation means choosing a small k over a large d: the resistance d/k is what matters, and it adds like electrical resistance.

Typical values: copper 400 {W,m^(-1),K^(-1)}, glass 1, water 0.6, still air 0.026. The tiny conductivity of air is why foam, wool and double glazing work — they trap air so that convection cannot short-circuit the conduction path.

3. Convection

A fluid heated from below becomes less dense, rises, cools, and sinks: a **convection current**. The rate depends on geometry and on whether the flow is laminar or turbulent, and it is not captured by a simple formula in the way conduction is. The physics is worth remembering qualitatively:

- Convection is much more efficient than conduction in a fluid, so a forced draught (a fan) can carry away heat far faster than still air.
- Heated rooms stratify, with warm air at the ceiling — the opposite of what a well-designed radiator position should give.
- In the Earth's mantle, convection drives plate tectonics; in the Sun's interior, it carries the energy that eventually leaves as sunlight.

4. Thermal radiation

Every body above absolute zero radiates. The power emitted from area A at temperature T is

$$P = \varepsilon\sigma\, AT^4,\ \sigma = 5.67\times10^{-8}\ \{W,m^{-2},K^{-4}\}, \tag{2}$$

where ε is the **emissivity** (0ε1; a perfect black body has ε=1).

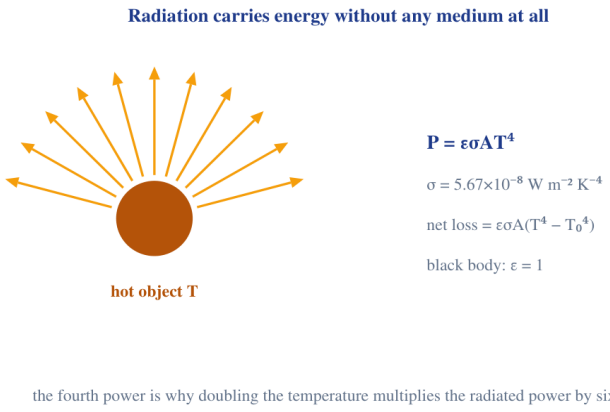


Figure 2. Radiation is electromagnetic, so it needs no medium: the Sun warms the Earth through a vacuum. A body both emits and absorbs, and the net exchange is the difference of the two fourth powers.

A body both emits and absorbs, so its net loss to surroundings at T_0 is

$$P_{\{net\}} = \varepsilon\sigma\, A(T^4-T_0^4). \tag{3}$$

The T^4 makes radiation negligible at room temperature and dominant at high temperature — which is why a hot poker glows and a warm one does not. The wavelength at which the emission peaks obeys **Wien's law**:

$$\lambda_{max}T = 2.90\times10^{-3}\ \{m,K\}. \tag{4}$$

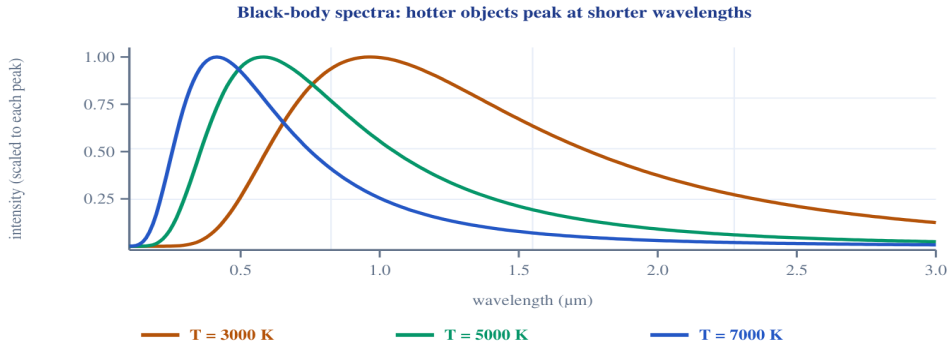


Figure. Wien's law λ_max T = 2.9×10■³ m K links the peak wavelength to the temperature.

A body at 300 {K} peaks at λ≈9.7 μ{m} — the infrared, which is why thermal cameras work. The Sun at 5800 {K} peaks at 500 {nm}, in the visible, and that is not a coincidence: eyes evolved to use the most abundant light available.

5. Specific heat and latent heat

Heating a body of mass m with specific heat capacity c takes

$$Q = mc\Delta T. \tag{5}$$

But changing the **phase** of a substance takes energy with no change in temperature:

$$Q = mL,\ L_f = 3.34\times10^5\ \{J,kg^{-1}\}\ (\{ice\}),\ L_v = 2.26\times10^6\ \{J,kg^{-1}\}\ (\{water\}). \tag{6}$$

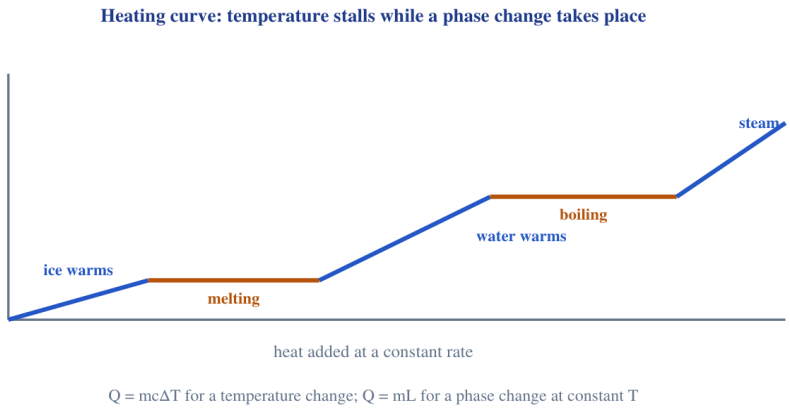
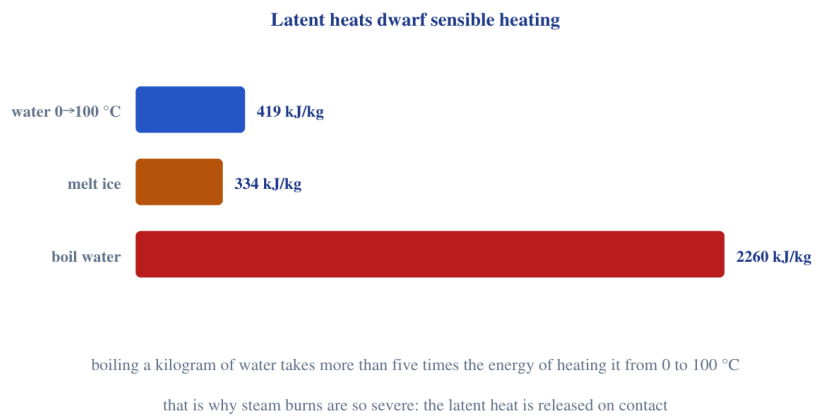


Figure 3. The plateaus are longer than the slopes by a large factor: melting a gram of ice takes as much energy as heating a gram of water through



$80^\circ\mathrm{C}$.

Figure 5. Phase c

which is why the energy is large and the temperature does not change during the change.

The latent heat of vaporisation of water is enormous: boiling away a kilogram takes more than five times the energy needed to heat it from 0 to 100°{C}. That single number explains sweating (evaporation cools the skin), the severity of steam burns, and why a pot of water takes so long to boil dry.

For a mixture or a calorimetry problem, the method is always the same: energy lost by the hot bodies equals energy gained by the cold bodies, **plus** any latent heat, and the final state may be partly melted — in which case the temperature is pinned at the melting point and you must check the energy budget before assuming it.

6. Newton's law of cooling

For small temperature differences, the rate of heat loss is proportional to the excess temperature:

$$(dT)/(dt) = -k(T-T_0) T = T_0 + (T_({start})-T_0)e^(-kt). \quad (7)$$

Interactive experiment: Newton’s law of cooling — open the online handout to use it.

That is the same exponential as radioactive decay, and for the same reason: the loss rate is proportional to the quantity that remains. The constant k lumps together conduction, convection and (weakly) radiation; when the object is much hotter than its surroundings the T^4 law dominates and the cooling is no longer exponential.

The heat-transfer checklist. Which mechanism? (solid → conduction, fluid → convection, vacuum or high T → radiation) → for steady conduction, treat it as a resistance network → for radiation, use P = εσ AT^4 and remember the net exchange needs a **difference** → for heating problems, separate the temperature changes (mcΔ T) from the phase changes (mL) and check whether the energy budget actually reaches the next phase → for cooling, use the exponential only for a small excess temperature.

7. Problems: reason, calculate, check

Ten original problems in three tiers: **[1]** foundation, **[2]** multi-step olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. State assumptions, keep units, and estimate the answer before computing it. Every problem has a

worked solution with a consistency check.

[1] Problem 1 — Heating water. A kettle contains 1.5 {kg} of water at 20°{C}; c = 4200 {J,kg^(-1),K^(-1)}. (a) Find the energy needed to bring it to 100°{C}. (b) Find the time if the kettle is rated at 2.0 {kW} and all the energy goes into the water. (c) Explain why the real time is longer.

Worked solution →

[1] Problem 2 — Melting ice. 0.50 {kg} of ice at 0°{C} is added to 2.0 {kg} of water at 25°{C} in an insulated container. (a) Find the final temperature, assuming all the ice melts. (b) Check that assumption by finding whether the water has enough energy to melt all the ice. (c) Find the final temperature if only 0.30 {kg} of ice were used instead.

Worked solution →

[2] Problem 3 — Conduction through a window. A glass window is 1.2 {m}×1.5 {m}, 4.0 {mm} thick, with k = 0.80 {W,m^(-1),K^(-1)}. (a) Find the heat current when the inside is at 20°{C} and the outside at 0°{C}. (b) Find the heat lost in a day, in kWh. (c) Find the thickness of a layer of still air (k = 0.026) that would give the same resistance, and comment.

Worked solution →

[2] Problem 4 — Radiation from a person. A person has skin area 1.8 {m²}, skin temperature 33°{C} and emissivity 0.97, in a room at 20°{C}. (a) Find the power radiated. (b) Find the net radiative loss. (c) Explain how the body balances this loss, and why you feel cold in a room with cold walls even when the air is warm.

Worked solution →

[2] Problem 5 — The Sun and the Earth. The Sun's surface is at 5800 {K} and its radius is 6.96×10^8 {m}; treat it as a black body. (a) Find the power radiated. (b) Find the intensity at the Earth, 1.5×10^(11) {m} away. (c) Find the equilibrium temperature of a black sphere at the Earth's distance, and explain why the real mean temperature is about 15°{C} rather than your answer.

Worked solution →

[2] Problem 6 — Newton's law of cooling. A cup of coffee at 90°{C} cools to 70°{C} in 5.0 {min} in a room at 20°{C}. (a) Find the cooling constant k. (b) Find the temperature after a further 10 {min}. (c) Find the time for the coffee to reach 30°{C}, and explain why the model eventually fails.

Worked solution →

[2] Problem 7 — Two layers of insulation. A wall consists of 10 {cm} of brick (k = 0.84) and 10 {cm} of insulation (k = 0.040), with inside at 20°{C} and outside at 0°{C}. (a) Find the heat flux through the wall. (b) Find the temperature at the interface between the layers. (c) Explain why the interface is much closer to the outside temperature.

Worked solution →

[3] Problem 8 — A calorimetry puzzle. A 0.20 {kg} block of ice at -10°{C} is dropped into 1.0 {kg} of water at 30°{C} in an insulated container. Take c_({ice}) = 2100, c_({water}) = 4200 and L_f = 3.34×10^5 {J,kg^(-1)}. (a) Find the energy needed to warm the ice to 0°{C} and then melt it. (b) Find the energy the water can supply cooling to 0°{C}. (c) Find the final state of the mixture.

Worked solution →

[3] Problem 9 — The greenhouse effect. The Earth absorbs solar radiation and re-emits it as a black body. (a) Show that a simple energy balance gives T = (S(1-α)/(4σ))^(1/4) for a planet with solar constant S and albedo α. (b) Evaluate it for S = 1360 {W,m^(-2)} and α = 0.30. (c) Explain why the observed mean surface temperature is about 33 {K} higher, and what a simple single-layer atmosphere model predicts.

Worked solution →

[3] Problem 10 — Design of a hot-water tank. A cylindrical tank of height 1.5 {m}, radius 0.35 {m} and insulation 8.0 {cm} thick (k = 0.035) holds water at 60°{C} in a room at 20°{C}. (a) Find the total surface area. (b) Find the heat loss rate. (c) Find the temperature drop per day for 200 {kg} of water, and explain the effect of the metal walls and pipe connections.

Worked solution →

Further practice and provenance

The 10 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this IPhO syllabus topic. They are not copied from official papers and are not official marking schemes.

Sources and scope. Topic coverage follows the IPhO syllabus (accepted 2014, amended 2015) as published by the IPhO organisation. Official problems are available through the IPhO past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified. Synthetic data, when used, are labelled as teaching data.

Where to learn more

IPhO syllabus and statutes

The coverage checklist this curriculum follows: accepted 2014, amended 2015. Read it once for scope, not for depth.

IPhO past problems and solutions

The only honest test of readiness. Work recent theoretical and experimental papers under time, then compare with the official marking schemes.

MIT OpenCourseWare 8.01/8.02 thermodynamics notes

Kinetic theory, the first and second laws, and heat transfer worked examples.